

CS617 – Monroe College: Week 3 Assignment exercises: Using R

In this exercise, you'll run the code in the child mortality script. This script includes all of the statements for the analysis presented in this chapter. As you run each statement, be sure that you understand what it does even if you're not yet comfortable coding it yourself.

This exercise replicates something that's likely to come up whenever you work in a team: being given an R script for an analysis, confirming you get the same results as your team member, and performing a few extra calculations or creating a new visualization.

View the data in the spreadsheet (optional)

1. If you have spreadsheet software such as Excel installed on your computer, use it to open this Excel file:
`Documents/murach_r/data/child_mortality_rates.xls`
2. Note that the spreadsheet contains five columns of data, one for the year and four for the age groups. Then, scroll down to view all of the data.

Open RStudio and use it to set the working directory

3. Start RStudio.
4. Open the script named `exercise_3-1` that's in this directory:
`Documents/murach_r/exercises/ch03`
Note that the comments make it easier for you to follow what the script does even if you don't yet understand all of the code.
5. Set the working directory to the same directory that contains the script. To do that, refer to figure 3-2. You can use the `setwd()` function, or you can select the appropriate item from the RStudio menus.
6. In the Console pane, enter the `getwd()` function to check the working directory.

Run the statements in the script

7. Confirm you have correctly installed the required packages and set the working directory by running the first four statements in the script. These statements load the packages you need, read the child mortality data, and display it.

Note: If you get a warning message that says, “Error: `path` does not exist”, you need to make sure that the Excel file exists and that the working directory is set to the same directory as the script.

8. Run the rest of the script statement by statement. For in-depth explanations of how each statement works, you can refer to the appropriate figure from this chapter.

Tip: While in the Console pane, you can use the up arrow to display and re-run statements. If you do that, remember that each statement runs using the variables currently in memory.

9. After you run the `saveRDS()` function, check to make sure the `mortality_long.rds` file has been saved to the current working directory.
10. Note that when statements are coded on multiple lines, each line typically ends with commas or operators like the addition (+) or pipe (%>%) operators.

Write and run some statements of your own

11. At the end of the script add a statement that uses the `head()` function to view only the first 3 rows of the `mortality_data` tibble. Then, add a statement that uses `tail()` to display the last 8 rows of the `mortality_long` tibble.
12. Display the summary data for the fifth column (15-19 years) of the `mortality_data` tibble.
13. Rename the four age group columns in `mortality_data` with the following names `Years1_4`, `Years5_9`, `Years10_14`, and `Years15_19`. This makes it easier to perform the calculation in the next step.
14. Use the `mutate()` function to add a new column to `mortality_data` named `AllAges` that adds the rate for each age group.
15. Create a line plot of the `AllAges` column. The `aes()` function should look like this:

```
aes(x = Year, y = AllAges)
```